

Course: Data Science Boot Camp — Part-Time
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Problem Statement

Some properties have the same location and description, but their rental prices are very different. In the Aribnb dataset collected on July 14, 2026, 1558 listings in Downtown with available price data had nightly prices ranging from \$7.96 to \$5277.6, representing a prices range of \$5269.64. This happens because hosts use different pricing strategies, change prices based on demand or season, or set different minimum stay rules. These price differences can create unusual numbers (outliers) that make calculating average prices and revenues less accurate.

Objectives

03

Q1: "Why is there such a significant disparity in rental prices, even though the properties are in the same area and have the same specifications?"

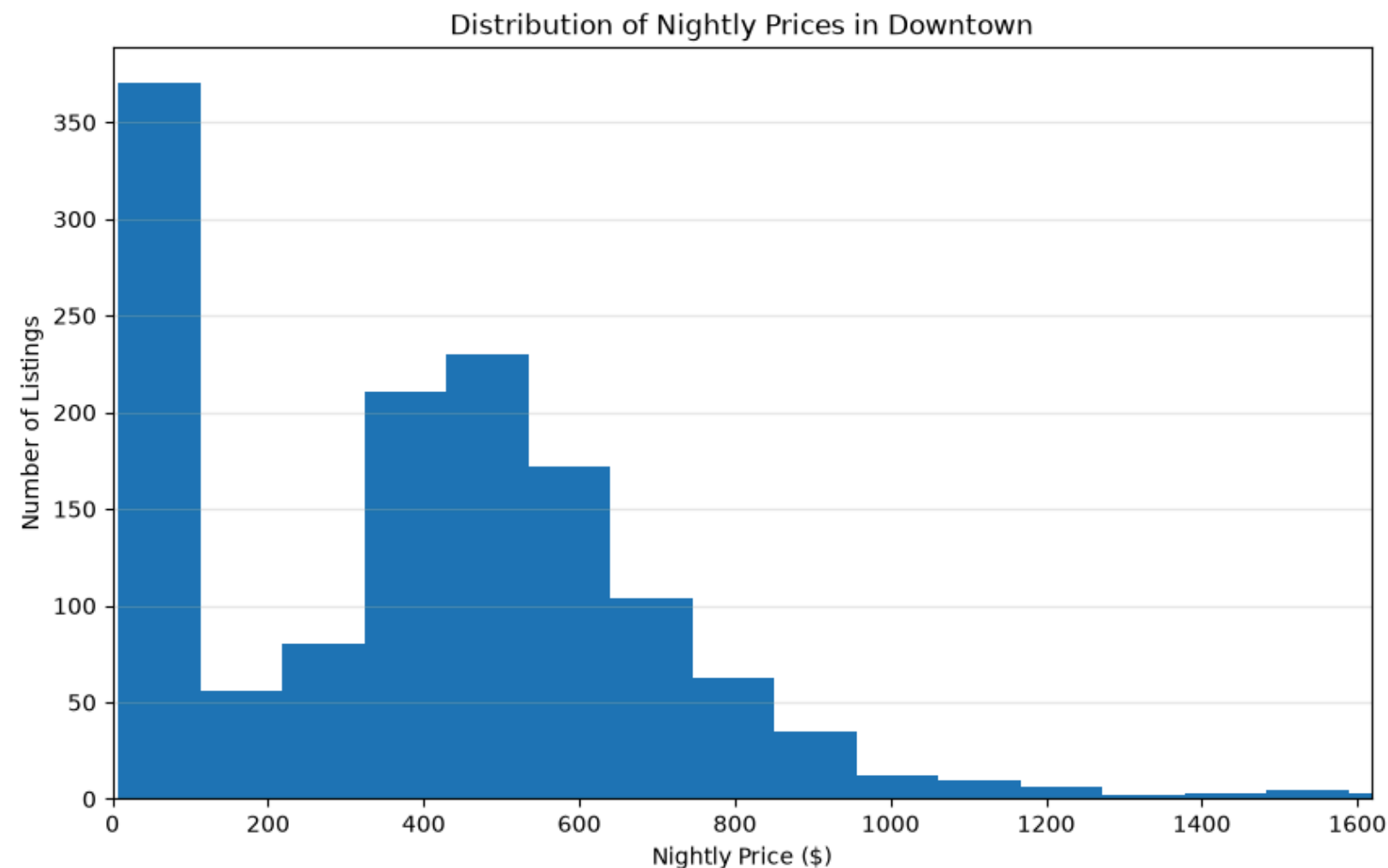
Q2: What is the unique factor that draws people to certain high-priced rentals, even when cheaper options are available?

Q3 : How significantly are customer feedback, listing liquidity, and minimum-stay rules correlated with nightly price premiums in Downtown?

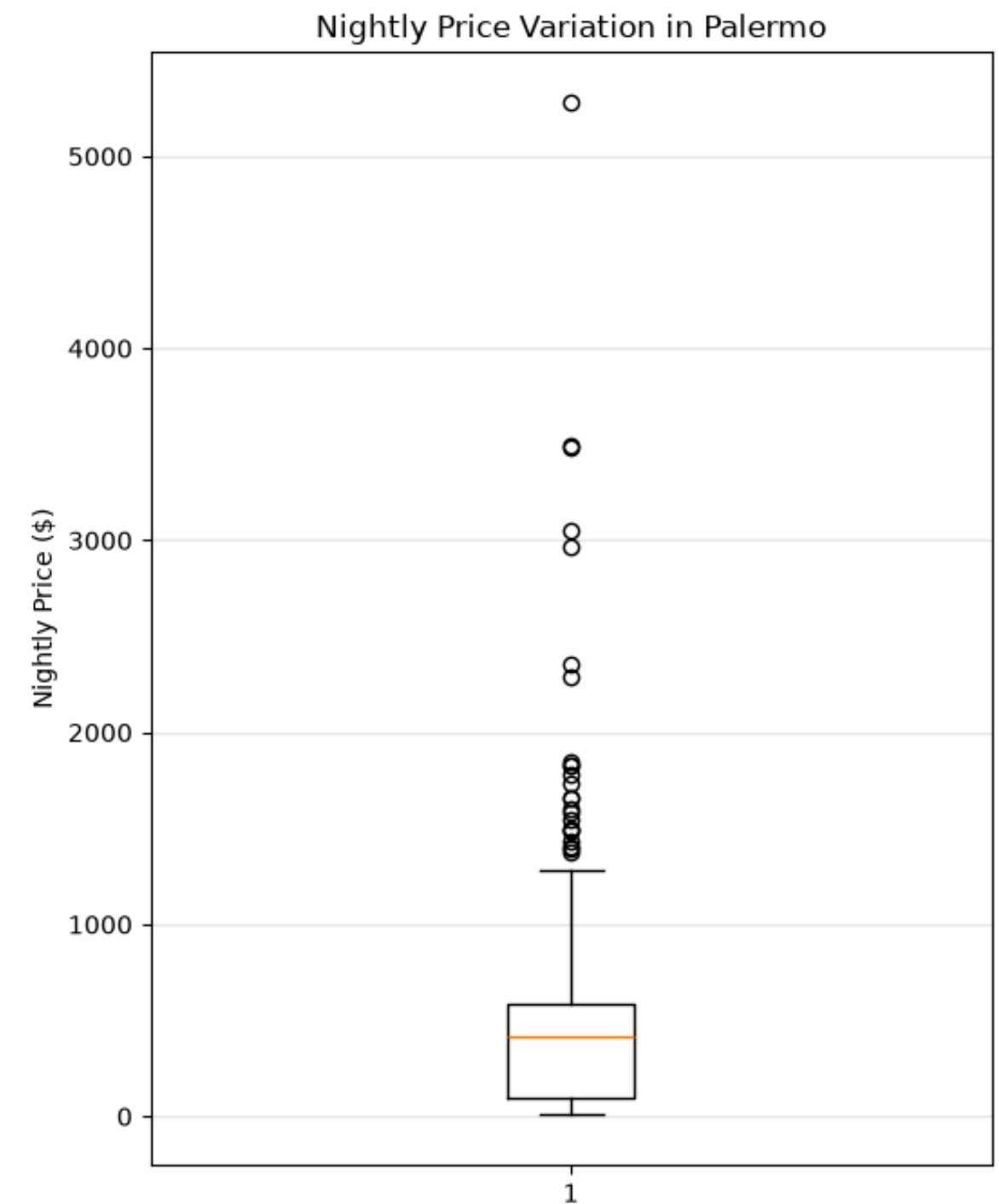
Objectives (Visualization)

03

Q1: "Why is there such a significant disparity in rental prices, even though the properties are in the same neighbourhood and have the same specifications?"



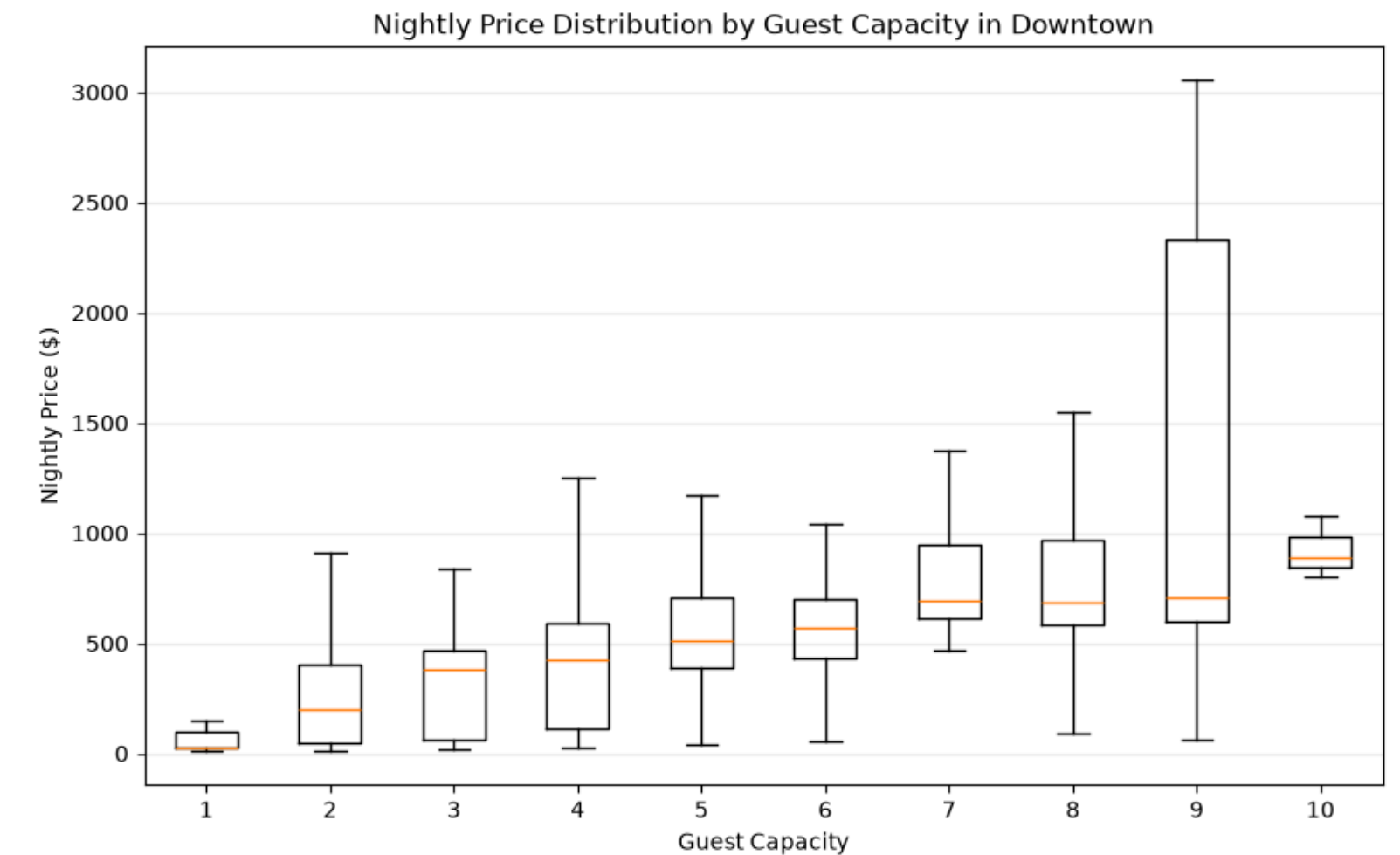
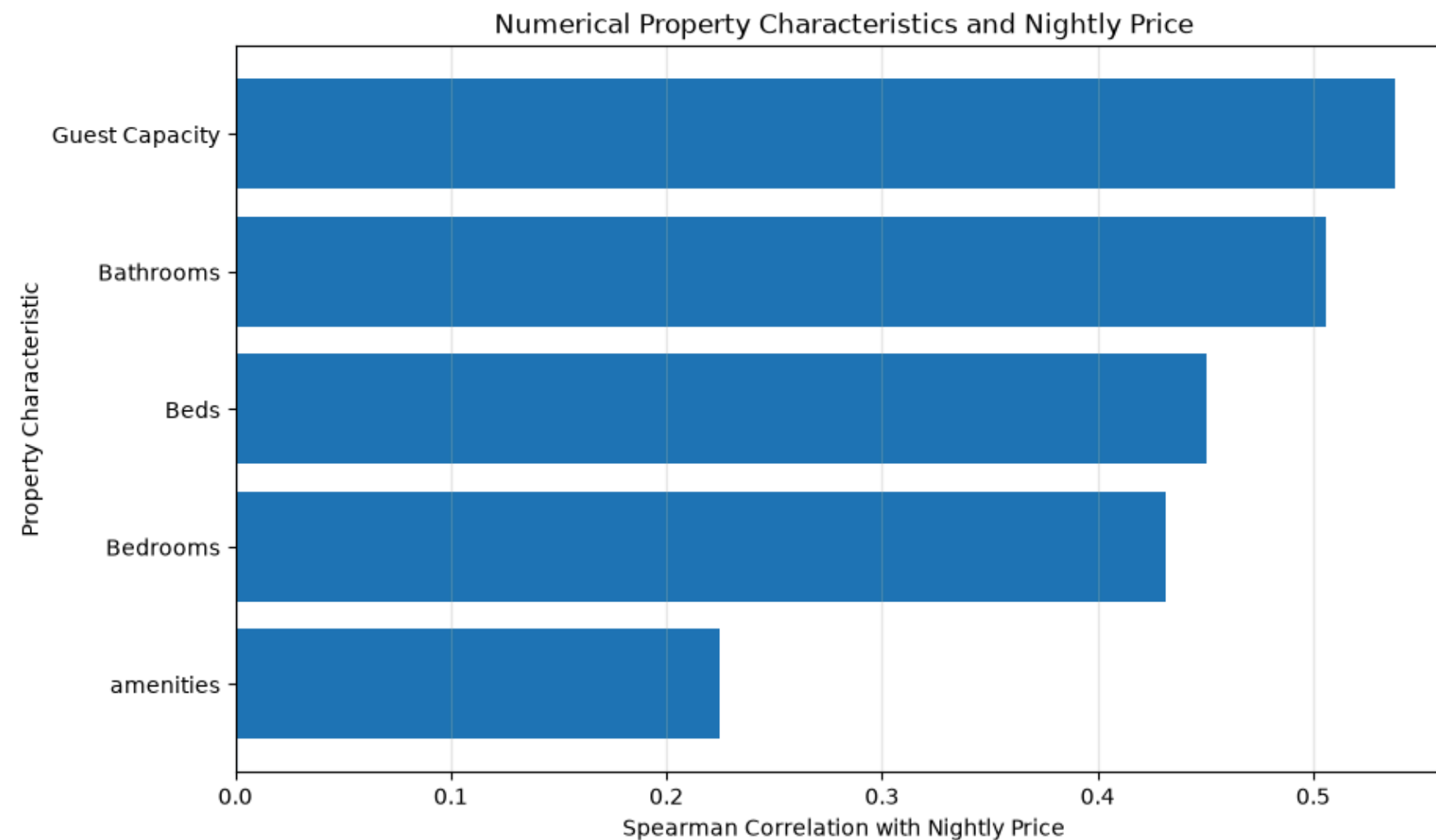
About 64.92 % of Downtown listings have nightly prices of \$500 or less, while a smaller percentage have higher prices. Overall, prices range from \$7.96 to \$5277.6 , giving a price range of \$5269.64.



Objectives (Visualization)

03

Q2: What is the unique factor that draws people to certain high-priced rentals, even when cheaper options are available?

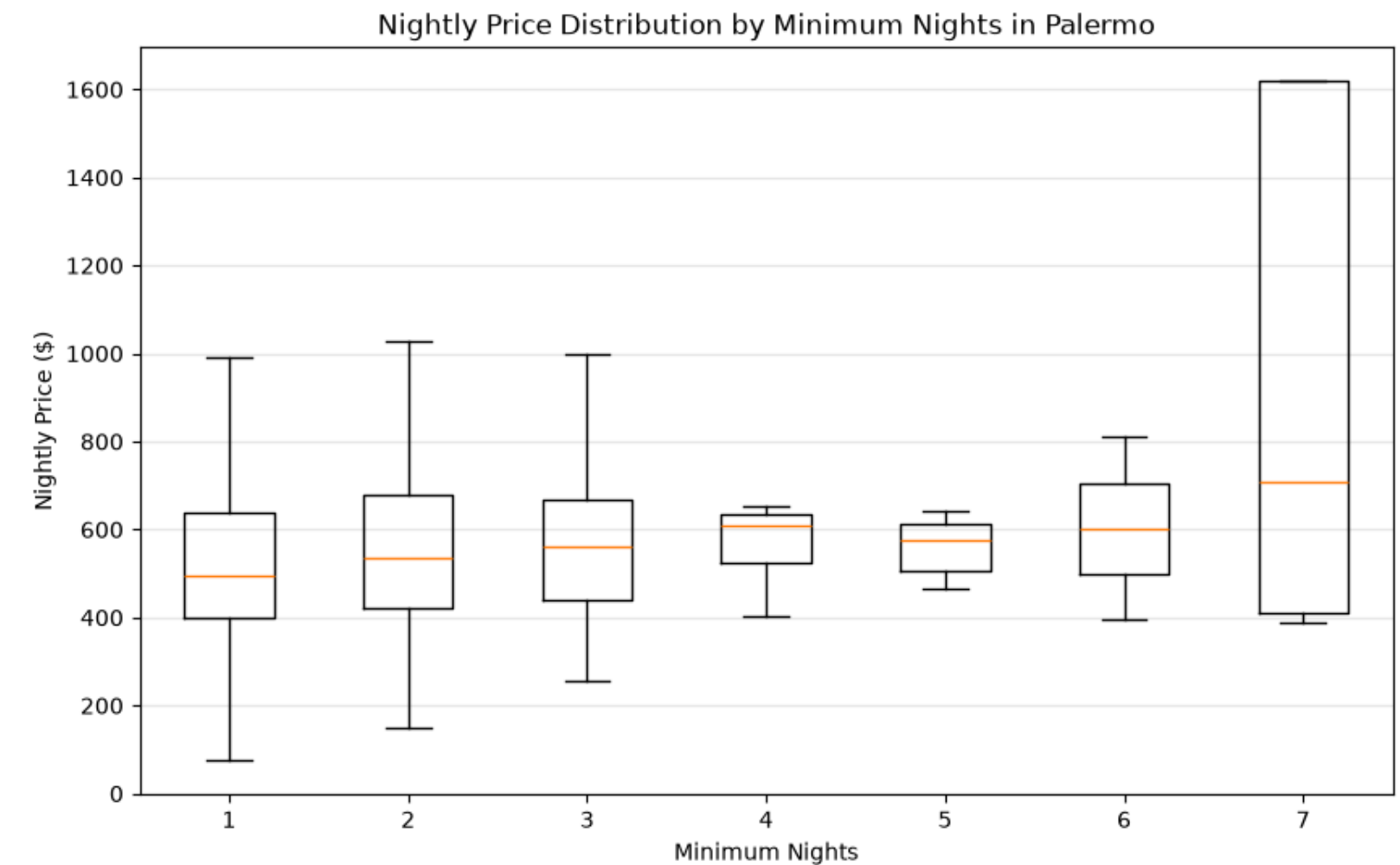
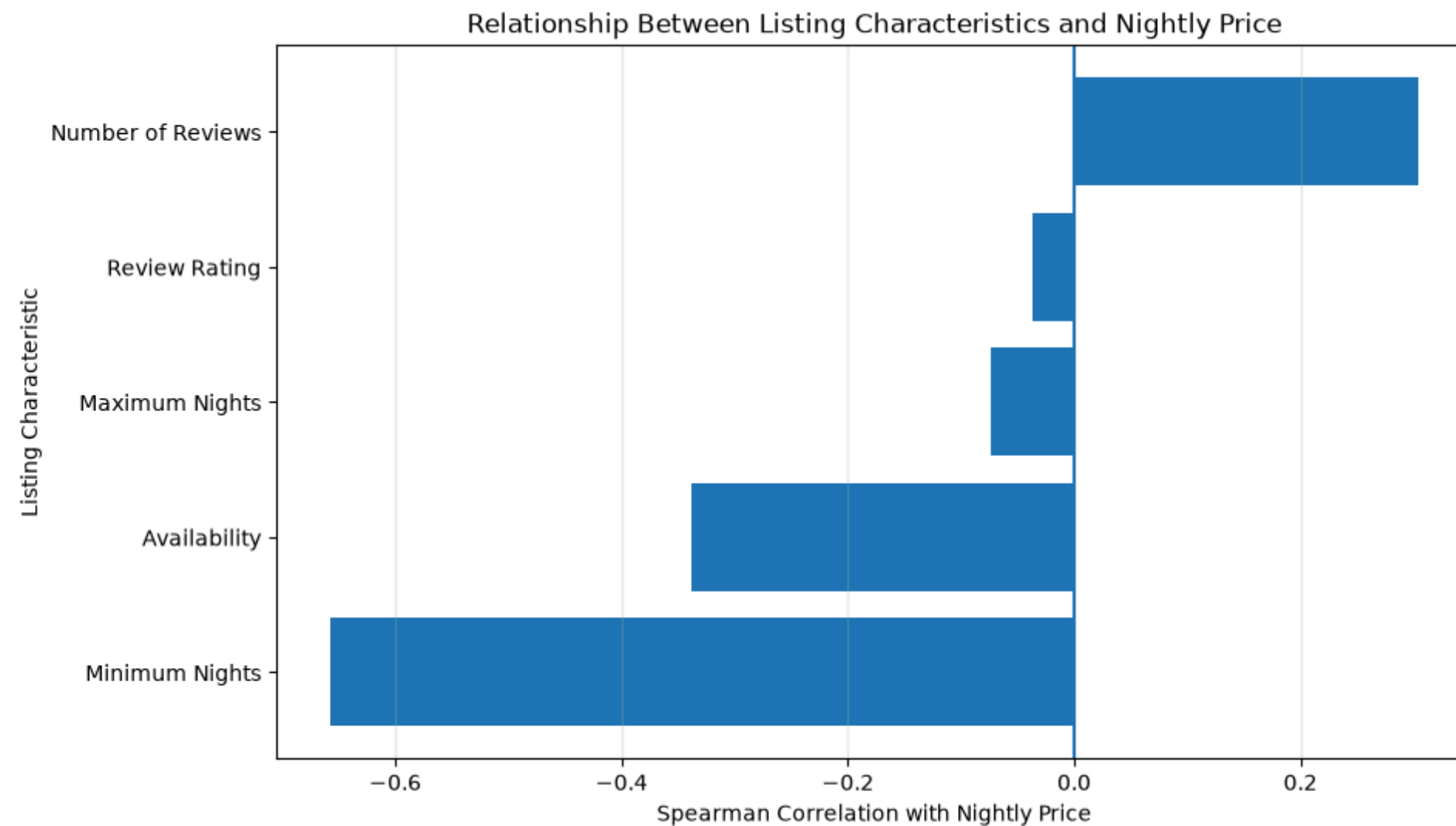


A positive relationship is observed between guest capacity and median nightly rates, demonstrating that listings accommodating larger groups command higher baseline prices.

Objectives (Visualization)

03

Q3: How significantly are customer feedback, listing liquidity, and minimum-stay rules correlated with nightly price premiums in Downtown?



Minimum nights showed the strongest negative relationship with nightly price, with a Spearman correlation of -0.65 . Listings with higher minimum-night requirements tended to have lower nightly prices.

Recommendations

1. Compare with similar nearby listings

Hosts should compare their nightly prices with nearby listings that have similar property type, room type, capacity, bedrooms, bathrooms, and amenities rather than relying only on neighbourhood averages.

2. Consider property characteristics when setting prices

Hosts should consider differences in property characteristics when evaluating whether their nightly price is competitive with nearby listings.

3. Review large price differences

Listings with substantial price differences from similar nearby properties should be reviewed to determine which characteristics may explain the difference.

4. Support pricing decisions with data

Airbnb hosts can use the identified factors associated with nightly price to make more informed and consistent pricing decisions.

Limitations

04

1. Quoted Price, Not Actual Payment

The dataset shows the listed nightly price, which may differ from the final amount actually paid by guests.

2. Unobserved Location Factors

Latitude and longitude identify location, but they do not capture factors such as views, noise, street quality, or proximity to specific attractions.

3. Missing Data

Some listing characteristics, ratings, and review-related variables contain missing values, which may limit comparisons between listings.